

# ***SuperKontrol 2.0***

Control software for the SuperK EXTREME series

## Software Manual



Issue: 3.1

Published: July 2012

Author: TOF/KNI

Copyright © 2012 by NKT Photonics A/S. All rights reserved. Reproduction or translations of any part of this work is prohibited.

## Table of Contents

|       |                                           |    |
|-------|-------------------------------------------|----|
| 1     | General .....                             | 3  |
| 2     | Installing the software .....             | 4  |
| 2.1   | USB Interface.....                        | 4  |
| 2.2   | SuperKontrol 2.0 Installation .....       | 7  |
| 3     | Using the SuperKontrol 2.0 Software ..... | 11 |
| 3.1   | Starting Up .....                         | 11 |
| 3.2   | Front Panel .....                         | 12 |
| 3.3   | About .....                               | 15 |
| 3.3.1 | System Information .....                  | 15 |
| 3.3.2 | Upload Firmware.....                      | 16 |
| 3.3.3 | Log .....                                 | 16 |
| 3.3.4 | Reset Current Limit .....                 | 17 |
| 3.4   | Interlock .....                           | 18 |
| 3.5   | Function Selector .....                   | 19 |
| 3.5.1 | Rep. Rate.....                            | 19 |
| 3.5.2 | SuperK SELECT .....                       | 20 |
| 3.5.3 | SuperK VARIA .....                        | 23 |

# 1 General

## Introduction

Please take the necessary time to read this software manual, which contains important information about how to install and use the SuperKontrol 2.0 software together with a SuperK EXTREME system.

## Warning

The SuperK EXTREME laser system is a laser Class 4 device and as such potentially dangerous. Read the SuperK EXTREME Instruction manual before continuing.



*Figure 1-1: Laser classification labels*

## Support

Do you have any questions concerning this software, please do not hesitate to contact NKT Photonics at [support@nktphotonics.com](mailto:support@nktphotonics.com) or on phone number: +45 45 787 787 during office hours 9:00-16:00 Central European Time (not during holidays).

## Description

The SuperKontrol 2.0 software is capable of controlling the SuperK EXTREME products (S24x-xxx-xxx) on a PC running Microsoft Windows.

## 2 Installing the software

### Introduction

The SuperKontrol 2.0 software communicates with the SuperK EXTREME system via a standard USB connection.

This section includes two parts:

- Installing the USB Interface (See Section [USB Interface](#))
- Installing the SuperKontrol 2.0 Software (See Section [SuperKontrol Installation](#))

### Software Download

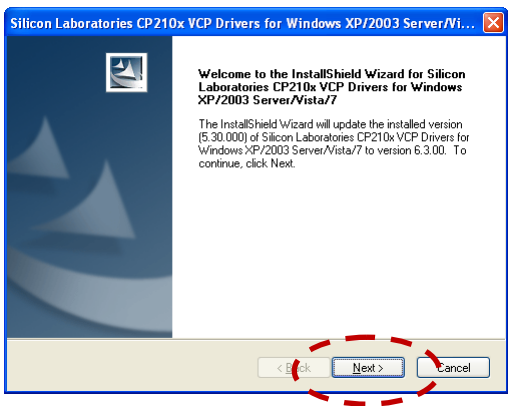
Please notice that the latest edition of the USB driver software and the SuperKontrol software may be downloaded from the NKT Photonics website. Go to: [www.nktphotonics.com/software](http://www.nktphotonics.com/software) or click on "Support" on the nktphotonics.com main page to go to the download section.

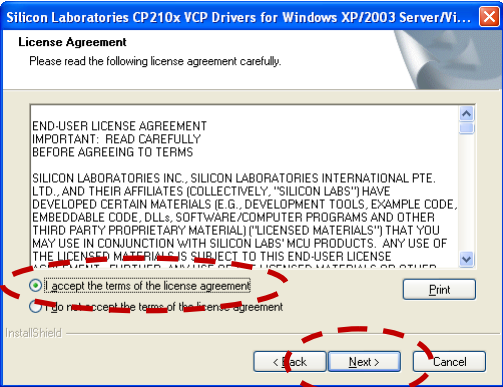
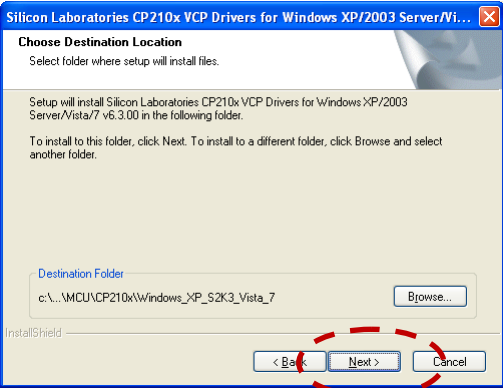
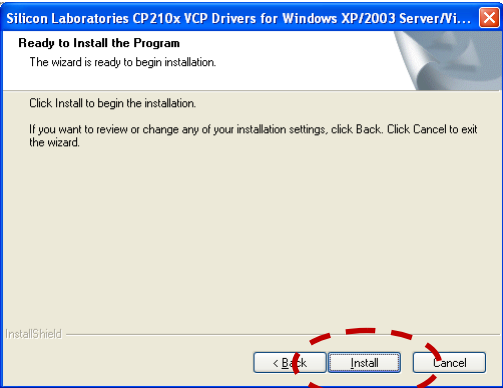
### 2.1 USB Interface

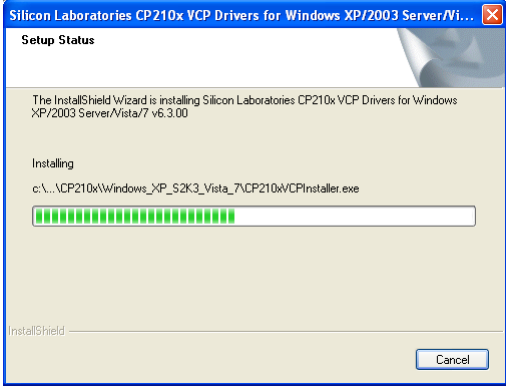
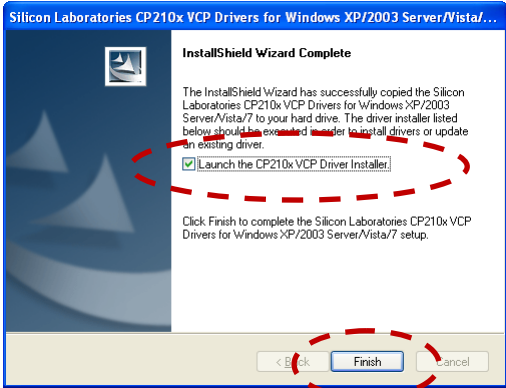
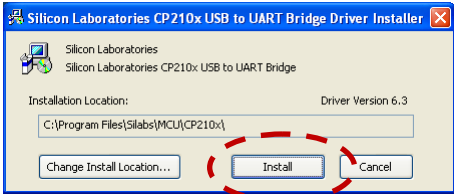
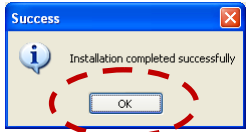
Install the Silicon Laboratories CP210x USB to UART Bridge software before connecting the system with the USB port on the computer.

### USB Driver

The following procedure describes how to install the driver software.

| Step | Description                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------|
| 1    | Place the SuperKontrol 2.0 Software CD in the CD-ROM drive or download software from nktphotonics web site.                    |
| 2    | Run CP210x_VCP_Win_XP_S2K3_Vista_7.exe from "D:\CP210x VCP driver", where D is the letter for the CD-ROM drive.                |
| 3    | Click Next on the window that appears.<br> |

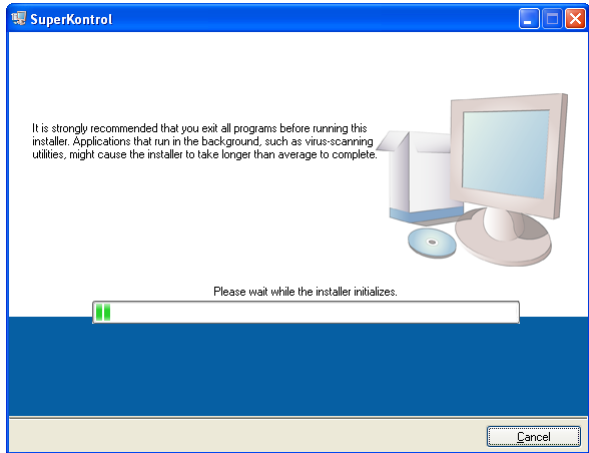
|   |                                                                                                                                                                                           |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <p>Read the License Agreement, choose I accept the terms of the license agreement, and click Next.</p>  |
| 5 | <p>Click on Next to proceed with the installation.</p>                                                 |
| 6 | <p>Click on Install to begin the installation.</p>                                                    |

|    |                                                                                                                                                                                                    |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7  | <p>Files are being copied to the computer.</p>                                                                   |
| 8  | <p>Select Launch the CP210x VCP Driver Installer and click Finish to begin the actual driver installation.</p>  |
| 9  | <p>In the dialog box that appears, click on the Install button to install the USB driver.</p>                  |
| 10 | <p>When the installation has completed, click OK to close the installer.</p>                                   |

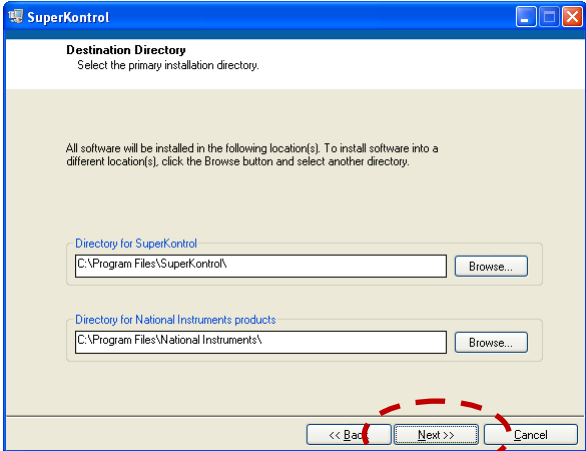
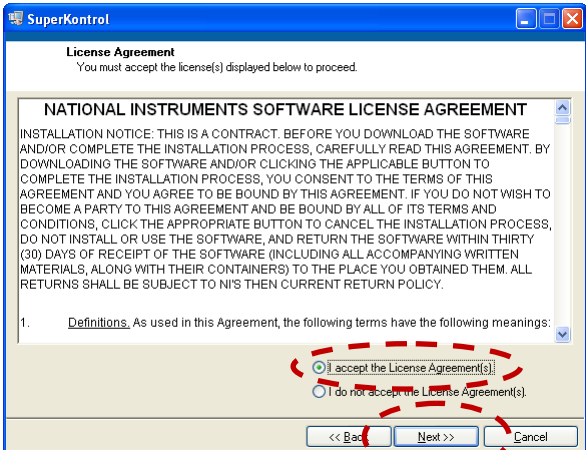
## 2.2 SuperKontrol 2.0 Installation

### SuperKontrol 2.0 Software

Use the following procedure to install the SuperKontrol 2.0 software.

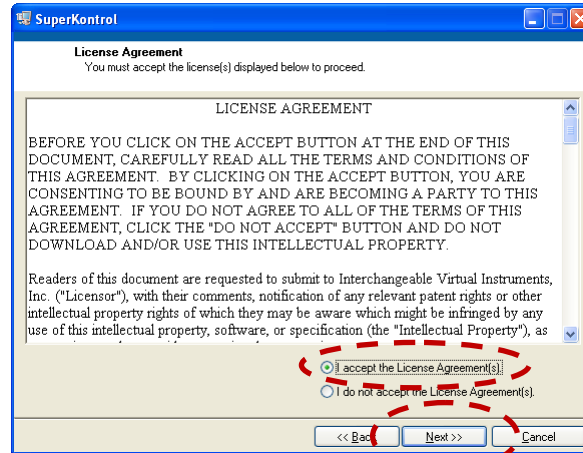
| Step | Description                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Place the SuperKontrol 2.0 CD in the CD-ROM drive or download software from nktpotonics web site.                                          |
| 2    | Run setup.exe.                                                                                                                             |
| 3    | <p>An installer window will show up on the screen.</p>  |



|   |                                                                                                                                                                                              |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <p>The installer asks for destination directories. Choose the suggested directories by clicking Next.</p>  |
| 5 | <p>Read the License Agreement, and choose I accept the License Agreement(s) and Next to proceed.</p>      |

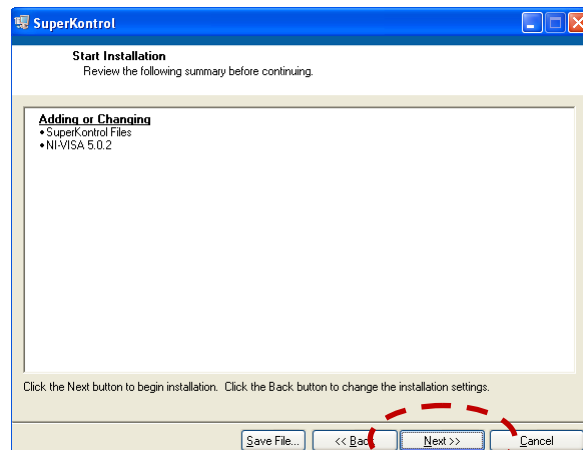
6

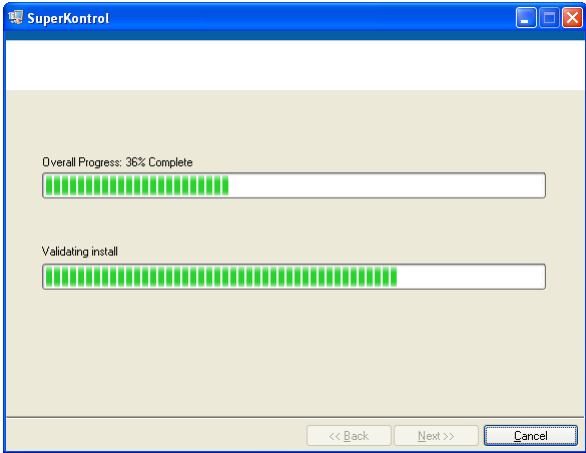
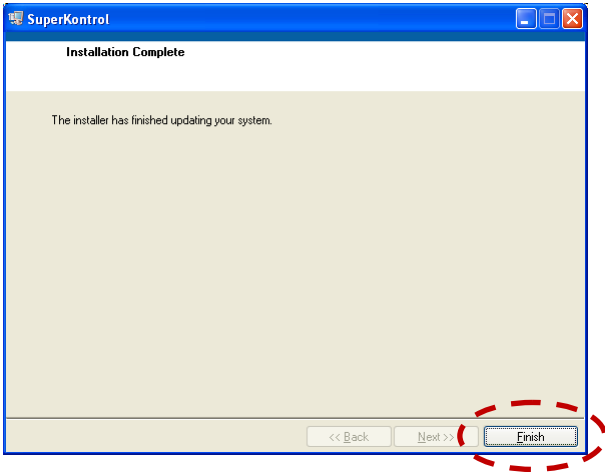
Read the License Agreement, and choose I accept the License Agreement(s) and Next to proceed.



7

Click Next to start copying files to the computer.



|    |                                                                                                                                                                                                                                 |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8  | <p>A new window shows the progress of the installation.</p>                                                                                   |
| 9  | <p>When the installation has completed, click Next to close the installer.</p>                                                               |
| 10 | <p>On the desktop and under Start, Programs you will find shortcuts to SuperKontrol 2.0. Click on one of them to run SuperKontrol 2.0.</p>  |

## 3 Using the SuperKontrol 2.0 Software

### Introduction

This section includes 5 parts, representing the overall control of the user interface:

- Starting Up: How to start up the SuperK Software.
- Front Panel : General introduction to the user interface.
- About: System information and download of log file.
- Interlock : Information about interlock states.
- Function Selector: Advanced control of the SuperK EXTREME system.

### 3.1 Starting Up

#### Start

Start SuperKontrol 2.0 via Start - Programs – SuperKontrol or with the SuperKontrol short-cut on the desktop.

During SuperKontrol 2.0 start-up it will automatically scan for accessible communication (COM) ports that may be used for communication with a SuperK EXTREME system. If more SuperK EXTREME systems are connected to the same computer, the software will try to establish communication at the lowest COM port number. If communication cannot be achieved at one COM port it will automatically try the next possible port until communication has been established.

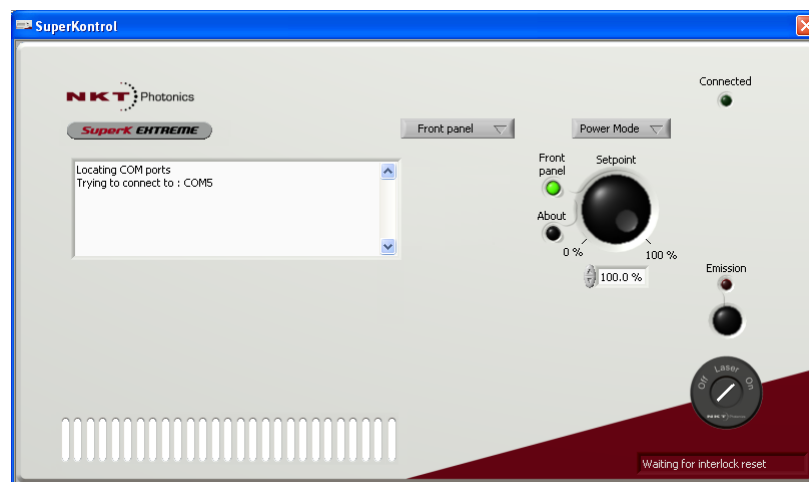


Figure 3-1: SuperKontrol software interface

If the USB connection has not been provided between the computer and the SuperK EXTREME system the software will inform that no accessible communication port has been identified.

If the USB connection is established but the SuperK EXTREME system is not turned on, the SuperKontrol 2.0 program will find the COM port but not be able to connect to it.

When SuperKontrol 2.0 establish communication with the SuperK EXTREME system, it will change its front panel to show the actual display of the system – unless an error code is present, which instead will be shown.

### 3.2 Front Panel

The SuperKontrol 2.0 front panel looks similar to the front panel of the SuperK EXTREME system. The window size is 800x450 pixels, which makes it possible to view the full window on e.g. a small laptop computer with screen resolution at 800x600 pixels.

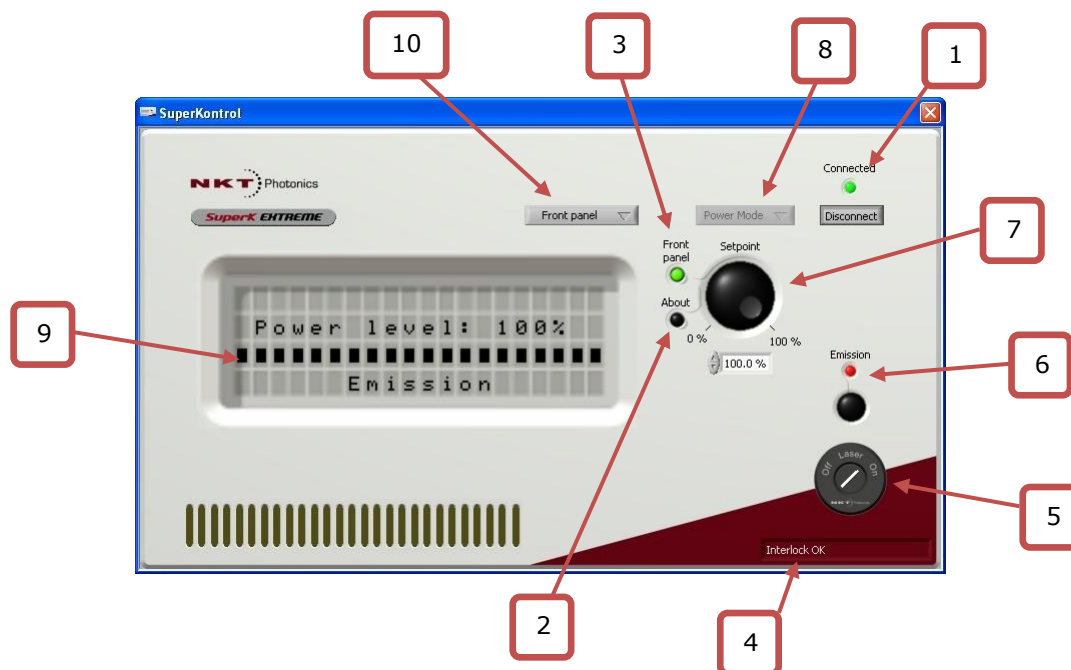


Figure 3-2: SuperKontrol interface when connected to the SuperK Extreme system

#### 1 - Connect and Disconnect

When communication has been established between computer and SuperK EXTREME system the green Connected LED will emit. To disconnect the computer from the SuperK EXTREME system and to terminate the SuperKontrol 2.0 software click on the Disconnect button. If the connection is lost the Connected LED will turn off and the Connect button should be clicked to establish connection again.

If the LED emits red, this indicates a communication problem, e.g. external accessories with same addresses. To fix this, please check the rotary switches on the external accessories and ensure that all accessories connected to the External Bus have their individual addresses.

#### 2 - About

Click on the About button to find out more about the system and the software being used and to download log data from the SuperK EXTREME system. Read the [About](#) section for more information about this.

#### 3 - Front panel

Click on the Front panel button to read out the front panel from the actual SuperK EXTREME system.

- 4 - Interlock** This field provides information about the interlock status. After e.g. a power cycle the interlock circuit needs to be reset in order to initiate laser emission from the SuperK EXTREME system. An interlock reset is either provided by clicking on the return/back button on the SuperK EXTREME system or on the Interlock (4) field in the SuperKontrol 2.0 software. Read section [Interlock](#) for more information about interlock.
- 5 - Key Switch** The key switch indicates the state of the actual key switch on the front plate of the SuperK EXTREME system. To enable laser emission the key switch must be in the "ON" position.
- 6 - Emission** When SuperKontrol 2.0 is communicating with the SuperK EXTREME system, the red LED lights up when the SuperK EXTREME system is emitting light. Click on the black button below the red LED to toggle On/Off laser emission.
- 7 - Turning Knob** The output power from the SuperK EXTREME system is adjusted via the turning knob or the associating field just below the turning knob. Coarse adjustments are easily made with the turning knob. Fine adjustments can be made with the Page Up, Page Down, ↑ and ↓ buttons on the computer keyboard. Page Up and Page Down adjust the setpoint in steps of 1 %, whereas ↑ and ↓ adjust the setpoint in steps of 0.1 %. A desired level in 0.1 % resolution can be typed in directly in the field below the turning knob.
- 8 - Control Mode** On the Control Mode selector the control mode of the SuperK EXTREME system can be changed between constant current or constant power. In constant current mode the current floating in the booster pumps is held constant, whereas the output power of the system is held constant in power mode. It is only possible to shift between the two control modes when laser emission is off.
- If the SuperK EXTREME system features the External Modulation option, two additional modes becomes available: External Current Mode and External Power Mode. In External Current Mode the system is operated in current mode, but can be controlled with the Modulation Input on the back of the SuperK EXTREME system. Similar in External Power Mode the system is operated in power mode, but can be controlled with the Modulation Input on the back of the system. Common for the two modes is that the maximum output level is set with the turning knob (7).
- If the SuperK EXTREME system features the External Feedback option, an additional mode becomes available: External Feedback Mode. In External Feedback Mode the system will if possible maintain a stable feedback level on the External Feedback input. In External Feedback Mode the setpoint can be adjusted in the range from 0 to 90 %.

Please note that even if there are connectors for External Feedback and Modulation Input on the back of the SuperK EXTREME system, the options are necessarily not available in the system. The External Feedback and Modulation Input are add-ons to the SuperK EXTREME system.

#### 9 - Display

The display area is used for many different purposes.

If the About (2) button is clicked the display area will provide information about hardware and software being used.

If the Front Panel (3) button is clicked the display area will provide information about what is written on the actual display on the SuperK EXTREME.

If the Function Selector (10) button is clicked, the display area will provide information about other things dependant on options available with the actual system and associating accessories.

#### 10- Function Selector

Click on the Function Selector (10) for more advanced control of the SuperK EXTREME system. Read section [Function Selector](#) for more information about this.

Select Rep. Rate to obtain information about the repetition rate of the light from the SuperK EXTREME source. For systems containing the Pulse-Picker option, it is furthermore possible to adjust the repetition rate.

## 3.3 About

### 3.3.1 System Information

#### System Information

In the About window it is possible to obtain information about what revision of SuperKontrol 2.0 is being used and at what COM port number the communication between computer and SuperK EXTREME system is established.

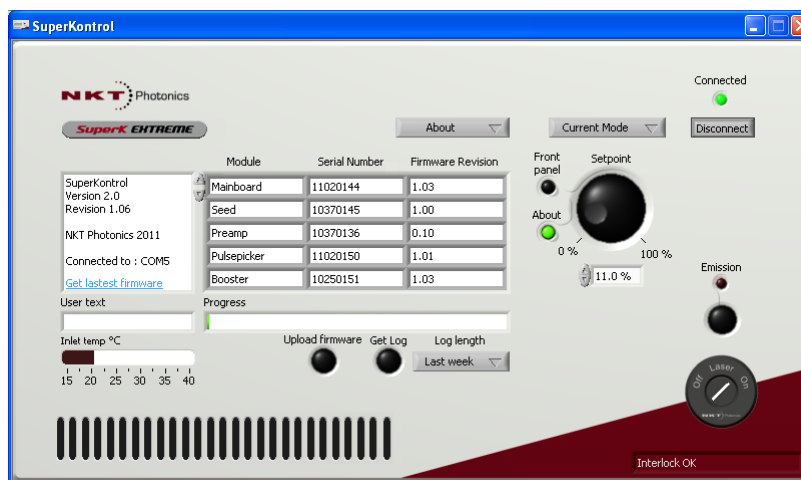


Figure 3-3: The "About" menu

The array in the center of the front panel contains information about modules available in the SuperK EXTREME system and external accessories connected to the External Bus interface.

A SuperK EXTREME system consists of a:

- Mainboard
- Seed Module
- Pre-Amp Module
- Pulse-Picker (optional)
- Booster Module
- Front Panel

Accessories connected to the External Bus are listed in addition to the modules listed above. With the up and down arrows in the upper-left corner of the table it is possible to scroll down and up.

For each module the serial number and firmware revision is listed.

#### User Text

Text typed into this field will appear in the top line of the display in the SuperK EXTREME system. This User text can be used e.g. to identify a system in case multiple systems are available.

#### Inlet Temperature

The SuperK EXTREME system is cooled by air flow forced by the internal fans. The temperature of the air inlet is monitored by the Inlet temperature sensor.

#### Get Latest Firmware

Click on the blue text "Get latest firmware" to open an Internet browser window with NKT Photonics homepage where the latest firmware package can be downloaded from.



### 3.3.2 Upload Firmware

#### Upload Firmware

With SuperKontrol 2.0 it is possible to upload firmware to the SuperK EXTREME system. The firmware can be uploaded to a single module or to the complete system dependant on the HEX-file to be uploaded.

Click on the Upload firmware button, locate and choose the HEX-file with the new firmware to be uploaded. The progress and the estimated remaining time for firmware upload is shown on the progress bar.

### 3.3.3 Log

#### Log

The SuperK EXTREME system continuously saves data to an internal storage. The logged data can be downloaded from the system to the computer by SuperKontrol 2.0. The log file is saved in a format, which can only be read by NKT Photonics e.g. as a part of a service check.

#### Emission Off

Please note that laser emission will automatically be turned off when a log download is initiated and it is not possible to turn it on again until the download has been completed.

#### Log Download

Choose the desired period to download from the drop down menu and click on Get log. Following a dialog box appears, where a target directory for the log file should be chosen. Click on Current Folder to begin downloading of log files to the chosen directory.

The file name for the log file is automatically generated as:  
SN<System serial number>\_SysLog\_<Log period>\_<Date>.bin,  
where:

- <System serial number> is the 8 digit serial number of the SuperK EXTREME system.
- <Log period> is the chosen period of downloaded data.
- <Date> is the current date read from the SuperK EXTREME system (ddmmm\_yyyy).

In addition to the above log file, a system log file is downloaded. This text-file contains the module and firmware history of system. The file name for this system log is automatically generated as: SN<System serial number> SysLog <Date>.txt, where system serial number and date is as described above.

While log data is downloaded from the SuperK EXTREME system to the computer the progress bar is continuously updated.

The time it takes to download the log file to the computer mainly depends on the amount of data to be downloaded, which is determined by the time of operation. In the table below the download time is provided for the different Log periods to choose between assuming operation 24 hours per day 7 days per week.

| Log period    | Maximum download time |
|---------------|-----------------------|
| Last week     | ~ 6 minutes           |
| Last month    | ~ 26 minutes          |
| Last 3 months | ~ 78 minutes          |
| Last 6 months | ~ 156 minutes         |
| Full          | ~ 313 minutes/year    |

### 3.3.4 Reset Current Limit

If the SuperK EXTREME system observes too much reflected power it shuts off laser emission immediately and turns down the Current Limit in the system in steps of power at 20%. When a reflection occurs the SuperKontrol will shift to the Error window and the reflection will be shown as an error event.

When the SuperK EXTREME system has turned down its Current Limit, this can be reset with SuperKontrol. Click on the About button to get to the About window. Click here on the Reset Current Limit button to reset the limit.

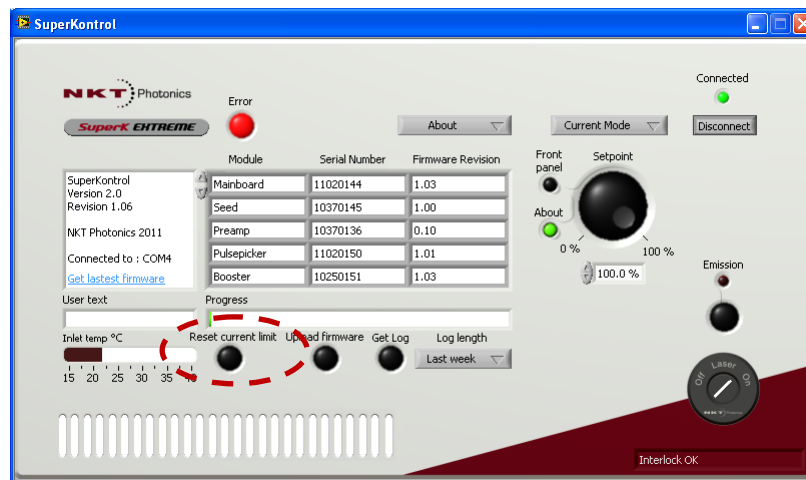


Figure 3-4: Reset of current limit

### 3.4 Interlock

The interlock circuit can obtain several states. In the table below each state is listed with a description of the state.

| State                              | Description                                                                                                                                                                                                                                                        |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waiting for interlock reset        | The system is waiting for the operator to reset the interlock circuit. Click on the Interlock (5) field to reset.                                                                                                                                                  |
| Interlock ok                       | The system is ready for use.                                                                                                                                                                                                                                       |
| Front panel interlock / key switch | Front panel key switch is in off position or the internal cover switch is open (during service).                                                                                                                                                                   |
| Door switch                        | The interlock connection on the back of the SuperK EXTREME system for a door switch is open.                                                                                                                                                                       |
| External module interlock          | Interlock is open on the External Bus interface on the back of the SuperK EXTREME system. If no accessories are connected to the SuperK EXTREME system a sub-D plug with a short-connection between pin #3 and #4 must be connected to the External Bus interface. |
| Application interlock              | The internal Application interlock connection inside the SuperK EXTREME system is open. Contact NKT Photonics if this state occurs.                                                                                                                                |
| Internal interlock                 | Interlock is open inside the SuperK EXTREME system. Contact NKT Photonics if this state occurs.                                                                                                                                                                    |
| Interlock power fail               | The voltage on the interlock signal is not as it should be. If this state occurs, then check if the interlock signal accidentally is connected to ground/case.                                                                                                     |
| MCU blocks interlock               | Mainboard micro-controller is not permitting interlock, relays to close e.g. during firmware upload.                                                                                                                                                               |
| Relay contact failure              | Failure on relay or key switch contacts. Contact NKT Photonics if this state occurs.                                                                                                                                                                               |

## 3.5 Function Selector

The Function Selector (#10 in figure 3-2) is intended for more advance control of the SuperK EXTREME system and control and monitoring of external accessories. The following sub-sections describe how to use these options.

### 3.5.1 Rep. Rate

#### With Pulse-Picker

On systems including the Pulse-Picker option, it is possible to adjust the repetition rate of the system. Click on the Function Selector and select Rep. Rate. You will get a relative view on the output signal and the NIM Pulse output as shown on the screen below.

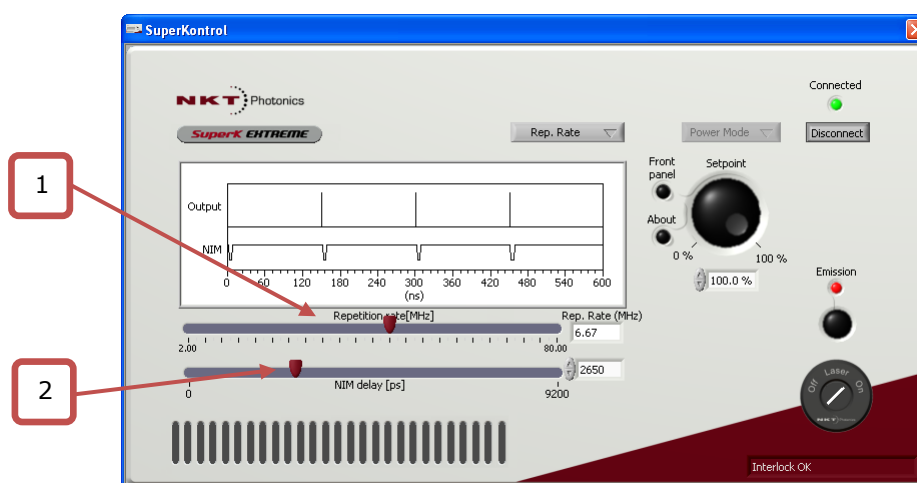


Figure 3-5: Rep.rate menu

#### 1 - Repetition Rate Adjustment

Click on the repetition rate division slider to adjust the repetition rate. Right to the repetition rate division slider the repetition rate is shown, either in frequency (MHz) or in time (ns). Click on the field to shift between frequency and time.

#### 2 - Trigger Adjustment

To adjust the delay of the NIM Pulse trigger signal, use the NIM delay slider below the Repetition rate division slider. The NIM Pulse trigger signal can be delayed up to 9200 ps. If a larger delay of the trigger signal is required it can be necessary to add a longer coax cable from the NIM Pulse output on the SuperK EXTREME system to where the trigger signal is used.

#### Without Pulse-Picker

For SuperK EXTREME system without the Pulse-Picker option the repetition rate cannot be changed and there is no NIM Pulse trigger output to delay. The repetition rate and the seed pulse output of the system is shown as described above.

### 3.5.2 SuperK SELECT

When a SuperK SELECT accessory is connected to the External Bus and one or two associating RF-drivers when the SuperK EXTREME system is powered up, the SuperK SELECT control window becomes available by choosing the SuperK SELECT option from the Function Selector.

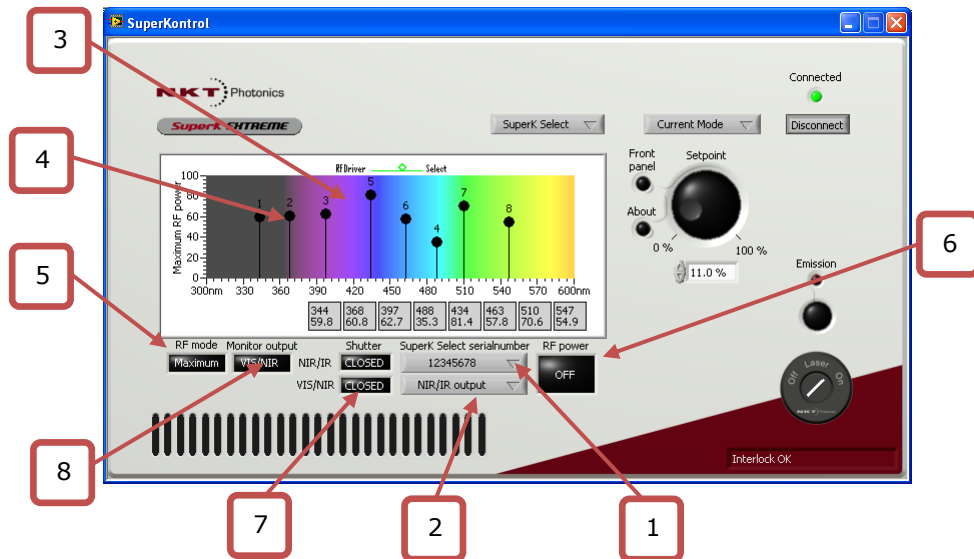


Figure 3-6: SuperK Select interface

1 - Unit to be Controlled

The system can be used with more than one SuperK SELECT connected to the External Bus. If multiple SuperK SELECTs are connected, choose the serial number of the SuperK SELECT unit to be controlled from the drop down window.

2 - Output to be Controlled

If the interfacing and chosen SuperK SELECT features two crystals it possible to shift between the VIS/NIR and NIR/IR output on the SuperK SELECT when the RF power is off. Click the drop down window to choose between the two outputs.

3 - Color Scheme

Selecting an output with a crystal for visible wavelengths turns the 2-dimensional color scheme into a palette of available colors as shown on the example above. Wavelengths outside the visible range will be presented as grey, so if an infrared output is selected the color scheme will turn into grey.

4 - Channel Control

The RF-driver features 8 channels, i.e. an output with a crystal can be configured to up to 8 different wavelengths at the same time. When the SuperK EXTREME system is powered up all 8 channels are inactive.

In the lower-left corner of the color scheme drag one black dot at a time and drop it into the color scheme at the desired wavelength and power level. To maintain a certain wavelength but to control the power level, press SHIFT down on the keyboard, click on the channel (dot) on the color scheme and move it up or down.

If channels should be configured more accurately than possible with drag-and-drop, a dialog box for each channel can become

available either by pressing the CTRL key down on the keyboard and clicking on the desired channel (dot) with the mouse or clicking on the grey square boxes underneath the color scheme (channel 1-8 in sequence).

#### 5 - RF Mode

With the RF Mode button it is possible to shift between Optimum RF power and Maximum RF power.

With the system configured to operate up to Optimum RF power, the power level in the main peak increases for increasing RF power.

With the system configured to operate up to Maximum RF power, the main peak will increase until a certain level of RF power is reached where after increased RF power will only lead to an increase in side band power level and the power in the main peak will start decreasing.

Please refer to the SELECT manual for further information on choice of RF power.

#### 6 - RF Power

To activate the output from the SuperK SELECT the RF power must be turned on. Please notice that the coax cable between the RF driver and the SuperK SELECT should not be disconnected when the RF power is on.

#### 7 - Shutter

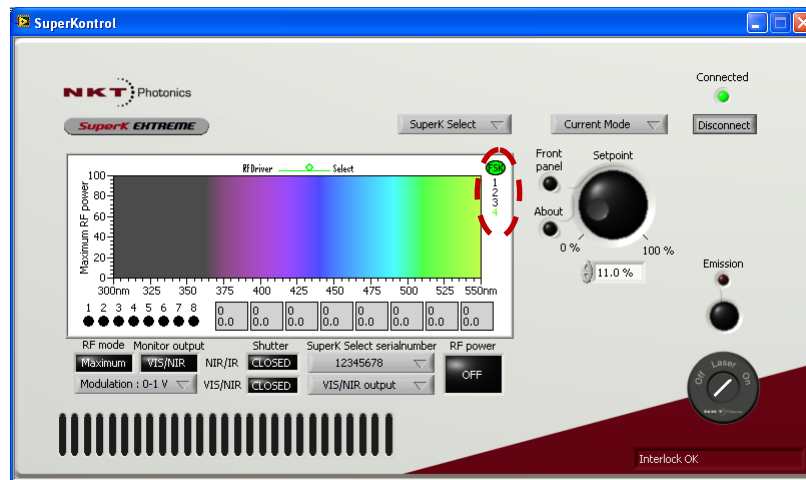
The SuperK SELECT has a mechanical shutter for both the VIS/NIR and NIR/IR output. The SuperKontrol 2.0 software makes it possible to view the state of these two shutters.

#### 8 - Monitor Output

Together with the SuperK SELECT there is an Option to use the External Power Monitor. An External Power Monitor can be connected to either the NIR/IR Monitor, VIS/NIR Monitor or both, With the Monitor Output button the user selects what External Power Monitor signal to pass through to the Monitor Output on the SuperK SELECT. The External Power Monitor can be used to feed a power signal back to the SuperK EXTREME system for power locking. The External Power Monitor can be purchased separately or in addition to the SuperK SELECT.

#### FSK

The SuperK SELECT has an Option for Frequency Shift Keying (FSK). See figure 3-7 for details.



*Figure 3-7: FSK option in the SuperK Select interface*

With the FSK Option available an FSK button is shown in the SuperK SELECT window. For details on using the FSK Option please refer to separate Application Note on this topic.

### 3.5.3 SuperK VARIA

When a SuperK VARIA accessory is connected to the External Bus, the SuperK VARIA control window becomes available by choosing the SuperK VARIA option from the Function Selector.

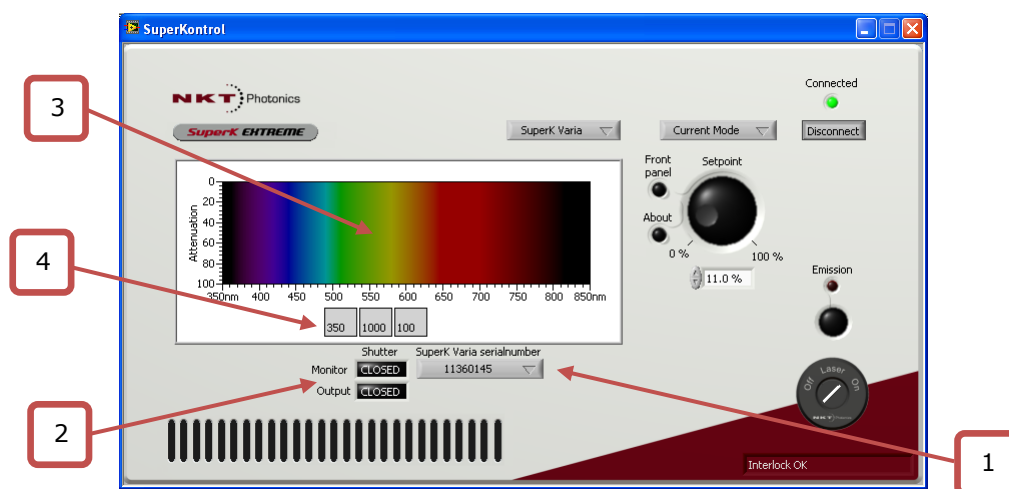


Figure 3-8: SuperK Varia interface

1 - Unit to be Controlled

The system can be used with more than one SuperK VARIA connected to the External Bus. If multiple SuperK VARIAs are connected, choose the serial number of the SuperK VARIA unit to be controlled from the drop down window.

2 - Shutter

The SuperK VARIA has a mechanical shutter for both the Monitor and Output. The SuperKontrol 2.0 software makes it possible to view the state of these two shutters.

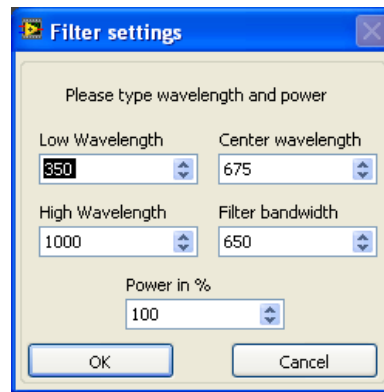
3 - Color Scheme

There are several possibilities to control the wavelength and output power from the SuperK VARIA. The chosen wavelength range and output amplitude is shown on the 2-dimensional color scheme. Output amplitude and wavelengths filtered out are greyed out.

4 - Filter Settings

Click on one of the three buttons underneath the color scheme to get to the Filter settings window. From here the Low and High wavelengths can be set or as an alternative the Center wavelength and Bandwidth can be set. Power amplitude can be set if the filter is present.





*Figure 3-9: Filter settings window when controlling the SuperK Varia*

#### Drag and Drop

With the mouse it is possible to drag and drop filter settings by clicking on the wavelength or amplitude settings and move it to new positions.

**NKT Photonics**

Blokken 84

3460 Birkerød

Denmark

Phone: +4543483900

Fax: +4543483901

[www.nktphotonics.com](http://www.nktphotonics.com)